

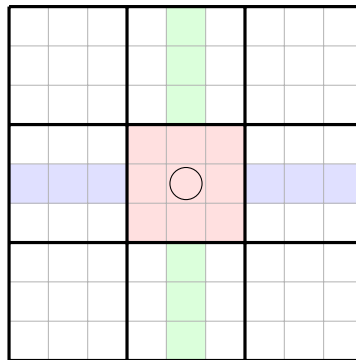
Sudoku

1 The puzzle and its rules

A sudoku grid is a 9×9 board of cells, subdivided into nine 3×3 *boxes*. A puzzle starts with some cells filled in (the *givens*); the task is to fill every remaining cell with a digit from 1 to 9 such that

- every **row** contains each digit exactly once,
- every **column** contains each digit exactly once,
- every **box** contains each digit exactly once.

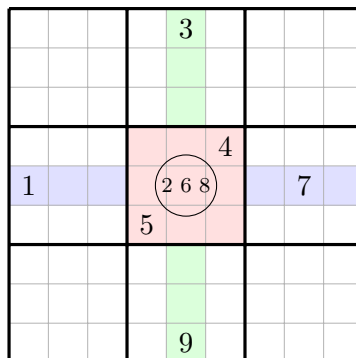
Each cell therefore belongs to three *units* (its row, its column, its box), and its 20 distinct *peers* — the other cells of those units — can never share its digit.



The three units of the circled cell: its row, its column, its box.

2 Candidates and constraint propagation

The solver's central data structure attaches to every empty cell its *candidate set*: the digits not yet ruled out. A digit placed in a cell is erased from the candidates of all 20 peers. In the position below, the circled cell sees 1, 7 in its row, 3, 9 in its column and 4, 5 in its box, leaving candidates $\{2, 6, 8\}$ — the small digits a pencil-and-paper solver would write in the corner:



Elimination: $\{1, \dots, 9\}$ minus the row, column, and box gives $\{2, 6, 8\}$.

Eliminations feed two deduction rules, and the rules feed further eliminations:

- **Naked single.** A cell whose candidate set shrinks to one digit must hold that digit. Placing it erases the digit from 20 more candidate sets — which may create new singles.
- **Hidden single.** If, within one unit, some digit remains possible in only one cell, that cell must hold it — even if the cell still has other candidates. Below, the digits 2, 4, 7 are missing from the row; 4 appears in only one candidate set, so it goes there:

1	2 4 7	3	5	2 7	6	2 7	8	9
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A hidden single: of the three unresolved cells, only the shaded one still has 4 among its candidates.

Running eliminations and both rules until nothing changes is *constraint propagation* — computing a fixpoint. Its power is easy to underestimate: most published easy and medium puzzles dissolve completely under propagation alone, with never a guess.

3 Search: when logic stalls

On hard puzzles propagation stalls short of a solution: a full pass over all the rules changes nothing — no elimination, no single — yet some cells still hold several candidates. The fixpoint has been reached, and it is not yet the answer; the rules at hand are exhausted. Here is a stalled row, and what a guess does to it:

1	2 7	3	5	2 4	6	4 7	8	9	stalled: 2, 4, 7 each have <i>two</i> possible homes
↓ guess 2 in the shaded cell, propagate									
1	2	3	5	4	6	7	8	9	the guess (orange) revives the cascade (green)

No single rule fires above — every unresolved cell has two candidates, every missing digit two homes. Guessing 2 strips 2 from the middle cell's candidates, leaving a naked single 4; placing it leaves 7 — the cascade runs again.

The solver then guesses: pick an unresolved cell, try one of its candidates, propagate, and recurse; a contradiction (some cell left with no candidates) refutes the guess, and backtracking tries the next one. This is the familiar search tree — with two twists that keep it tiny compared to the tours and queens of earlier quizzes:

- **Choose the most constrained cell** (fewest candidates remaining), the *minimum remaining values* rule — the same rescue-the-scarce principle as Warnsdorff's onward counts. A cell with two candidates gives a wrong guess a 50% chance of instant refutation.
- **Propagate after every guess.** Each guess triggers the full cascade, so consequences surface immediately and contradictions appear after a handful of placements, not deep in the tree.

4 Facts worth knowing

- **Uniqueness.** A *proper* puzzle has exactly one solution; verifying properness means searching for a second solution and failing. A puzzle generator must run exactly that check on every candidate puzzle it digs out.

- **Seventeen givens.** No proper 9x9 puzzle with fewer than 17 givens exists (McGuire, Tugemann and Civario, 2012 — by exhaustive computer search), and 17-given puzzles do exist.
- **It scales badly, provably.** On generalized $n^2 \times n^2$ boards, sudoku is NP-complete; the 9x9 case is only tamed by its fixed size and by how sharply propagation prunes.

5 From mathematics to program

The program keeps the grid as 81 candidate sets (a solved cell is a singleton) and implements propagation as the fixpoint of the two rules; the solver wraps it in MRV-guided backtracking. The renderer draws what solvers on paper actually do: pencil marks shrinking as eliminations cascade, deduced digits appearing in one color, guesses in another, and a contradiction flashing before its branch rewinds — constraint propagation made watchable.